HTML FUNDAMENTALS

Precise beginner-friendly notes • First-year level

HTML (HyperText Markup Language) is the standard markup language used to structure content on web pages. It describes what content is—such as headings, text, images, links and forms—not how complex behaviour works.

Learning outcome: After these notes, you should be able to read a basic HTML page, write common elements, create a form, use semantic tags and follow good coding habits.

1. Basics of HTML Structure

An HTML document is made of elements arranged in a tree-like structure called the **DOM (Document Object Model)**. A browser reads the document and displays its content.

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>My First Page</title> </head> <body>
<h1>Hello, world!</h1> <p>This is my first webpage.</p> </body> </html>
```

What each part means

Part	Purpose
<!DOCTYPE html>	Declares an HTML5 document; helps the browser use standards mode.
<html lang="en">	Root element; lang identifies the page language.
<head>	Holds metadata and links to resources; usually not visible in the page.
<meta charset="UTF-8">	Sets character encoding so a wide range of characters display correctly.
<meta name="viewport"...>	Helps the layout adapt to mobile screen widths.
<title>	Sets the browser-tab title and a useful page title for search results.
<body>	Contains the content displayed on the page.

2. Elements, Tags, and Attributes

A **tag** is the markup written inside angle brackets. An **element** usually consists of an opening tag, content and a closing tag. An **attribute** provides extra information in the opening tag.

```
<p class="intro">Welcome to HTML.</p> <!-- A void element: it has no closing tag --> <img
src="photo.jpg" alt="A mountain view">
```

In the example, ** ** and ** ** are tags; the complete paragraph is an element; **class="intro"** is an attribute. Common attributes include **id**, **class**, **href**, **src**, **alt**, **title** and **type**.

Void elements do not wrap content and do not have closing tags. Examples: <img>, <br>, <hr>, <input> and <meta>.

3. Headings, Paragraphs, Lists, Links and Images

Headings: Use <h1> through <h6> to create a logical hierarchy. Usually use one clear main page heading, then nest lower-level headings by topic. Do not choose heading levels just to change text size.

```
<h1>Web Development</h1> <h2>HTML</h2> <h3>Text Elements</h3> <p>HTML gives content its
structure.</p>
```

Lists: Use <ul> for unordered (bulleted) lists, <ol> for ordered (numbered) lists, and <li> for each item.

```
<ul> <li>HTML</li> <li>CSS</li> </ul> <ol> <li>Plan the page</li> <li>Write the markup</li> </ol>
```

Links: The <a> element creates a hyperlink. The href attribute gives its destination. Use meaningful link text so the purpose is clear.

```
<a href="https://example.com">Visit Example</a> <a href="about.html">About us</a>
```

Images: Use <img> with a valid src and descriptive alt text. The alt text communicates the image's meaning when it cannot be seen; use empty alt text (alt="") for purely decorative images.

```
<img src="campus.jpg" alt="Students walking across the campus" width="600">
```

4. Forms and Input Elements

Forms collect information from users. The <form> element groups controls. action identifies where data is sent and method specifies how it is sent. A server or suitable backend is needed to process submitted data.

```
<form action="/register" method="post"> <label for="name">Name</label> <input id="name" name="name"
type="text" required> <label for="email">Email</label> <input id="email" name="email" type="email"
required> <label for="course">Course</label> <select id="course" name="course"> <option
value="html">HTML</option> <option value="css">CSS</option> </select> <button
type="submit">Submit</button> </form>
```

Important controls and attributes

Element / attribute	Use
<input>	Single-line control; its behaviour depends on type.
type="text" / "email" / "password"	Text, email address, or masked password input.
type="radio" / "checkbox"	Choose one option from a group / choose independent options.
<textarea>	Multi-line text input.
<select> and <option>	A drop-down list and its choices.
<label for="...">	A text label linked to a control with the matching id.
name	Key used for the submitted form value; controls without a name are generally not submitted.
required	Asks the browser to require a value before submission.

GET vs POST: GET places submitted data in the URL and is commonly used for retrieval/search. POST sends data in the request body and is commonly used when submitting or changing data. POST alone does not encrypt data; use HTTPS for secure transport.

5. Semantic HTML

Semantic HTML uses elements that describe the meaning and role of their content. It makes pages easier to understand, maintain, navigate with assistive technology and interpret by search engines.

```
<body> <header>Site name and introductory content</header> <nav>Primary navigation links</nav>
<main> <article> <h1>News story</h1> <p>Article content...</p> </article> <section> <h2>Related
topics</h2> </section> </main> <aside>Related information</aside> <footer>Copyright and contact
details</footer> </body>
```

Common semantic elements

header = introductory content; **nav** = navigation links; **main** = page's main content; **section** = thematic grouping, usually with a heading; **article** = self-contained content; **aside** = related or supplementary content; **footer** = footer for a page or section; **figure** and **figcaption** = media/content with a caption.

Use and when no more meaningful element fits. They are generic containers, not semantic replacements for every element.

6. HTML5 New Features

HTML5 introduced a modern standard for web documents and several features that reduce the need for extra plugins. In current practice, HTML is maintained as a living standard, so “HTML5” is often used as a familiar name for modern HTML.

- **Semantic elements:** header, nav, main, section, article, aside and footer.
- **Audio and video:** native <audio> and <video> playback.

• **Graphics:** <canvas> for script-drawn graphics and SVG support for scalable vector graphics.
• **Forms:** input types such as email, date, number and range, plus built-in validation attributes.
• **Browser storage APIs:** localStorage and sessionStorage are web-platform APIs commonly used by modern web pages; they are not HTML tags.
• **Responsive support:** the viewport meta element helps pages fit mobile screens; CSS media queries handle responsive styling.

```
<video controls width="480"> <source src="lesson.mp4" type="video/mp4"> Your browser does not support video. </video> <audio controls> <source src="audio.mp3" type="audio/mpeg"> Your browser does not support audio. </audio>
```

7. HTML Best Practices

1. Start every page with <!DOCTYPE html> and set the correct lang attribute.

2. Use lowercase tag and attribute names; quote attribute values.

3. Indent nested elements consistently and keep the structure readable.

4. Close non-void elements and nest them correctly.

5. Use semantic elements and a logical heading order.

6. Give images meaningful alt text; provide labels for form controls.

7. Use unique id values; use class for reusable styling hooks.

8. Keep content, presentation and behaviour appropriately separated: HTML, CSS and JavaScript.

9. Use descriptive page titles and link text; avoid vague text such as “click here”.

10. Validate your markup and test pages in more than one browser and on mobile-sized screens.

11. Do not put passwords, secret keys or private data directly into HTML source.

Quick Revision

Question	Short answer
What is HTML?	A markup language that structures web content.
What is an element?	A complete HTML unit, often opening tag + content + closing tag.
What is an attribute?	Extra information written in an element's opening tag.
ul vs ol?	Bulleted list vs numbered list.
Why use alt?	To communicate an image's meaning when it cannot be seen.
Why semantic HTML?	Clearer structure, accessibility and maintainability.
Does HTML alone make a website interactive?	It structures content; CSS styles it and JavaScript adds behaviour.

Practice task: Create a one-page student profile using a page title, one h1 heading, two paragraphs, a list of three interests, one image with alt text, a link to a learning resource, and a form with name and email fields.

Reference tip: For authoritative element definitions and examples, use the MDN HTML documentation and the WHATWG HTML Living Standard.